Laboratory 6:
Conditional Image Generation

Computer Vision, Spring 2026

Overview

The goal of this session is to practice on some strategies for conditional image generation. Particularly, we consider: 1) the generation of artistic common images by using a simple deep-learning CNN approach and, 2) a face animation approach by using a novel GAN conditioning scheme based on Action-Unit (AU) annotations, which describe in a continuous manifold the anatomical facial movements defining a human expression. The approach permits controlling the magnitude of activation of each AU and combining several of them.

First of all, we must download the materials from the link *Laboratory 6 Materials* on Aula Global of *Computer Vision*.

FAQs:

- **What do I need to send?** A report with your work, including missing codes, an explanation where you discuss your work, as well as your results and conclusions. Extra code and analysis are welcome as well. However, it is worth noting that the provided .tex template must be used with no modifications. A maximum of 10 pages is allowed, excluding references. Your final score is not directly proportional to the number of pages in your report.
- **How?** Once your work is finished, you should upload it to the *Laboratory 6 Submission* on the Aula Global, in a single zip file. The submitted zip file has to contain your report and your source code (and all files needed to run it, such as images, other dependencies, etc.). **Be clear and concise.** The work in this session should be done in 2-people groups.
- **When?** Anytime until 8 June 2026 (23:59 UTC+2). After that, nothing will be received. **Within 24 hours prior to deadline, no doubts or comments will be answered.**

References

- Image generation slides of the course on *Computer Vision*.
- [1] *GANimation: one-shot anatomically consistent facial animation*, A. Pumarola, A. Agudo, A.M. Martinez, A. Sanfeliu and F. Moreno-Noguer. In International Journal of Computer Vision, 128(3): 698-713, 2020. Available on Aula Global of *Computer Vision*.
- [2] *A Neural Algorithm of Artistic Style*, L. A. Gatys, A- S. Ecker, M. Bethge. In Arxiv, 2015.

1 Artistic Style: Image generation with novel styles

In this task, we get familiar with a novel algorithm based on data to create artistic images of high perceptual quality. The algorithm combines content and style of arbitrary images, providing artistic images based on the content one and the style learned from the style image. To achieve that, the input image is represented as a set of filtered images at each processing stage in the Convolutional Neural Network (CNN).

- Tasks [M - Mandatory, O - Optional]:

  M Read the paper and try to understand the functionality of every module. You should report an explanation of every module in the learning algorithm. Explain the loss function and identify it in the code.

  M Run the function **main.m** for two arbitrary images. In the data folder you can find some pictures (dali.jpg, guernica.jpg, sunflower.jpg and, starryNight.jpg) to learn an artistic style. As a content image, the **main.m** function is considering the image lighthouse.png that you could use or change. A simple but realistic solution can be obtained by considering 50 iterations (more than that would improve the solution but increasing the computational cost, especially if you are using a commodity cpu for training). For gpu computation, between 1000 and 2000 iterations can be easily considered.

  M Provide the results and explain them. A qualitative evaluation of results after applying different styles is needed.

2 Facial Animation: Image generation with novel expressions

This task is to get familiar with GANimation, a modern deep-based approach for face animation from one single RGB image. To this end, you need to surf on the webpage https://github.com/albertpumarola/GANimation and follow the indications there are given. In particular, you should understand the modules provided there.

- Tasks [M - Mandatory, O - Optional]:

  M Read the paper and try to understand the functionality of every module. You should report an explanation of every module in the learning algorithm.

  M Follow the code and identify the cost function we use for training.

  M Following the paper, you can see a couple of failure cases. Could you imagine why that occurs?

  M From a theoretical point of view and considering the potential failure cases in the last point, how will you improve the algorithm to produce even more accurate solutions? Remember, the goal is to generate photo-realistic images (focusing on the face).

  O Try to train and test the code. As some computational resources are needed, to this end, you could consider some cloud servers such as Google Colab https://colab.research.google.com. A small fraction of the dataset can be considered for training.